

**RESEARCH ON GENDER PREDICTION FOR SOCIAL
MEDIA USER PROFILING USING MACHINE LEARNING
METHOD**

**Major Project Report Submitted to
SRI PADMAVATI MAHILA VISVAVIDYALAYAM**

In Partial fulfilment of the requirement for the

MASTER OF COMPUTER APPLICATIONS

IV SEMESTER

By

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Under the guidance of

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DEPARTMENT OF COMPUTER SCIENCE

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Tirupati-517502(A.P), Andhra Pradesh**

2021 - 2023

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CERTIFICATE

This is to certify that the project work entitled **“RESEARCH ON GENDER PREDICTION FOR SOCIAL MEDIA USER PROFILING BY MACHINE LEARNING METHOD”** is a bonafide record of work carried out by **BOLLIMUNTHA MEGHANA (ROLL NO: 2021MCA16011)** in the **Department of Computer Science, Sri Padmavati Mahila Visvavidyalayam, Tirupati** in partial fulfilment of the requirements of IV Semester of **MASTER OF COMPUTER APPLICATIONS**. The content of the Project Report has not been submitted to any other University / Institute for the award of any degree.

SIGNATURE OF THE GUIDE
Head of the Department

SIGNATURE

DECLARATION

I hereby declare that MCA IV Semester major project entitled “**Research on Gender Prediction for Social Media User Profiling by Machine Learning Method**” was done at the **Department of Computer Science, Sri Padmavati Mahila Visvavidyalayam**, Tirupati, in the year 2023 under the guidance of **Prof. T. Sudha** in partial fulfilment of requirements of MCA IV Semester.

I also declare that this project is my original contribution of the best of my knowledge and belief. I further declare that this work has not been submitted either in full or part for the award of any other degree of this or any other university.

Signature of the Student

ACKNOWLEDGEMENT

I am greatly indebted to our guide **Prof. T. Sudha** for taking keen interest on my project work and providing valuable suggestions in all the possible areas of improvement.

I express my sincere thanks to the teaching staff of the Department of Computer Science for extending support and encouragement to me in all the stages of the project work.

I gratefully acknowledge and express my gratitude to the non-teaching staff of the Computer Science Department who supported us in preparing the project report.

Signature of the Student

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ABSTRACT

This paper aims to propose a gender prediction method that integrates social media users' sentiment. This method is used to predict the gender attributes of users, so as to realize the research on the construction method of social media user portraits. Previous studies on gender prediction have done little analysis of sentiments. This paper has used the idea of transfer learning to analyze users' sentiments and integrated sentimental features into the existing machine learning, and hence, shows superior performance in terms of accuracy as compared to other methods. This paper mainly uses machine learning methods to realize the construction of social media user profiling. Gender attributes are studied. Firstly, feature extraction is carried out for text data of media users. Then the idea of transfer learning is used to analyze the user's sentiment and integrate the sentiment characteristics into the existing machine learning. Finally, five prediction methods, i.e., Logistic Regression (LR), Naïve Bayes (NB), k-Nearest Neighbor (KNN), Random Forest (RF) and Support Vector Machines (SVM) are used to predict the gender of the fused sentiments. The results show that the gender prediction effect after sentiment fusion is better than that before. and the accuracy is increased by about 2.1% on average.

INTRODUCTION

A Background

User profiling [1] is a tagging method based on the user's personal attributes, behavior habits and preferences. The construction of user profiling method [2] is the process of classifying users according to established categories through a series of data mining methods [3]. It can maximize the value and improve the team decision-making efficiency, so that users can improve the efficiency of information acquisition and accurately meet their own needs for product applications. The purpose of user profiling is to make a profiling of different dimensions of users and extract and label information such as users' demographic characteristics [4], social relations behavioral patterns, habit preferences and ideological views. In

recent years, with the rapid, large-scale and full-coverage development of the Internet, Sina Weibo (Social media line Twitter, Facebook, Sina Weibo) has become increasingly popular as a social network platform. It has the characteristics of large user group [5], fast news transmission speed, wide influence and group effect. Advertising media and social public opinion supervision departments urgently need to dig out accurate and usable information through the attribute analysis of Weibo users.

Related work

A few years ago, there was much research on the communication mechanism [6], marketing model and future development of social media users' profiling. For example, Liu Baoqin et al [7] constructed a SVM classifier to predict the gender of Weibo users by using the features of sentiment words and language style features related to sentiment. Dai Bin et al [8] used collaborative training algorithm to predict users' gender from different perspectives such as original microblog and forwarding microblog. Li ,et al [9] tried to predict the gender of interactive users by using the interactive text between sina weibo users. Wang Jingjing ,et al [10] predicted the gender of Weibo users by using the characteristics of Chinese microblog username and microblog text. AN Junhui [10] used Naive Bayes for gender recognition of Weibo username and verb respectively, but failed to integrate the two, which was limited and with low accuracy. HUANG Faliang et al [11] proposed to use K-nearest neighbor for gender prediction based on tolerance rough set method, but this method is susceptible to tolerance threshold. ZHANG Pu ,et al [12] used two methods, manual feature construction classifier and convolutional neural network model, which were used to predict the gender of users. The results of the two classifiers were fused with the XGBoost model to get the final prediction results. CAO Yang [13], for the issue that a single text data cannot accurately predict the gender of inactive users on Weibo, the base classifier was constructed by jointly learning the original microblog data, microblog username data and microblog source data, and then the base classifier was fused by Bayes to finally infer the gender of users. The above research did not take into account the sentimental differences of users of different genders, and users' sentiments can have an impact on user gender prediction.

Organization Profile :

Datapoint IT & Hardware Tech Pvt. Ltd.,

Business Proposition

Datapoint is incepted by young and ambitious team of Professional in the Industry with the Idea & motto of *“Simplifying Solutions & opportunities”*. Datapoint is into IT Training (Corporate/Individual), Project assistance, Software Development and Placements. Datapoint is one among the very few companies in Hyderabad, which are spread across all the areas and technologies. **Datapoint has been active in the profession of sourcing IT professionals from the year 2001.** We have since placed scores of candidates from different skill sets, with varying levels of experience.

Datapoint started its journey initially as a Consulting Company and as successful Placement Consultants as per the client's requirements we also emerged as a Corporate Training. Of-late we found that many engineering graduates are not being able to find jobs for themselves, despite increasing demand for IT professionals & even our clients couldn't be able to find suitable and potential candidates even among the freshers.

At this crucial point we found the gap which needed to be filled by Datapoint to improve our client satisfaction levels. The very decision of *“Training (IT & Non-IT aspects) & providing Project assistance”* to the freshers made Datapoint a significant player in the market. Datapoint is assisting many colleges and Organizations in Training & Recruiting freshers.

At Datapoint, unlike other training institutes, we know the industry requisites and what an organization expects from a candidate and henceforth we train our students accordingly so that they can get into the market with more confidence. Datapoint, as we already mentioned, not only trains extensively on technologies but also on soft skills. **Datapoint also motivates the students to implement the projects on their own, which gives them real time exposure towards the same.**

Datapoint endeavors to be a pioneer in Recruiting and manpower consulting thanks to strategic alliances with leading multinational companies in India and US of America. **Our technically competent, experienced, and certified consultants will help our clientele to get the right manpower at the right time.** We take pride in having top-notch companies who enable us to have faith in the future through maintaining high quality in screening, hiring and management.

Datapoint has identified several areas of thrust in the emerging and ever-growing IT industry and virtue of which, we would focus all our energies to get on to the fast track in the shortest possible period. We pursue requirements from leading Corporate in India and abroad.

Mission

“Our mission is to identify, recruit and facilitate quality manpower who are technically strong, dynamic and determined, as we are, for the future belongs to those who think and prove global.”

Why choose Datapoint for Academic Projects/Internship?

Our Project training is based on JAVA, DOTNET, EMBEDDED SYSTEMS, VLSI, MATLAB DSP/DIP, MATLAB SIMULINK as per industry expectations and we will allow the students to do the project in a real time environment under the guidance of industrial experts. We afford quality training to student which is evinced by the fact that several colleges recommended their students for our extensive project training

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We guarantee **reliable and productive candidates**, which is evinced by the fact that over **94% of our clientele** have done repeat business with us. We provide candidates at competitive bill rates, guaranteed not to change during the life of your assignment.

Once your positions are filled, they are there to stay.

“Simplifying Opportunities”: It’s more than just our slogan. Our entire business is geared toward helping our clients to successfully complete and implement their critical I.T initiatives in a timely and cost-efficient manner. Our job is to provide you with “Excellent manpower which makes the organization” so that you can be more successful and continue to be the leader in the industry.

Services:

Our Corporate Training Division has identified the following as thrust areas:

- IT Solutions
- Recruitment and Staffing
- Corporate Training
- Academic Projects (B.Tech/B. E (Mini & Major)/M.Tech/M.E/MCA), Polytechnic
- Internship
- Online Training

- CRT & Soft skills
- Workshops

For More Details, please reach us:

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Suresh Maguluri – CEO-9701043345, 7732074879

Datapoint IT & Hardware Tech Pvt Ltd.,

Why ?????

- Established in 2001.
- 22 Years of Expertise in Recruitments & Staffing.
- Having Own Software Development Division.
- Had a clientele of about 20 MNCs.
- The Company, which helps you to meet the industry expectations.
- Excellent track record in placing the Candidates of various levels.
- A company with the Coding standards of CMM level companies.

University Profile:

Sri Padmavati Mahila Visvavidyalayam (university for women) was founded in the year 1983 by N. T. Rama Rao, the Chief Minister of Andhra Pradesh, with the fervent desire to train women students as better builders of nation and to include skills of leadership in all aspects of life. The University was established under the Sri Padmavati Mahila Visvavidyalayam Act of 1983, which came into force on 14th of April 1983. It was started with ten faculties and 300 students and 20 staff members. In pursuance of objectives of university is awarded “A+ Grade “by NAAC.

The Campus of Sri Padmavati Mahila Visvavidyalayam is spread out in a lush green area of 138.43 acres. The university is situated at a distance of 3 kilometers from the railway and bus stations of Tirupati. The campus has the necessary buildings to run its academic programs and administrative machinery. There are separate Buildings for Humanities and Science, university’s Administration, Central Library, University Auditorium, Sericulture complex and school of Pharmaceutical Sciences and an independent building for Computer Science, Computer Center and Examination Hall.

The Objectives of the University:

- Emancipation of through Education.
- Acquisition of sound updated academic knowledge.
- Acquisition of specific vocational skills and competence for gainful occupation.
- Developing awareness of major issues and problems being faced by the society and by the women in particular.
- Acquisition of evaluation skills and training to function as better builders of the home and society.
- Acquisition of skills of leadership and entrepreneurship in all aspects of life. Keeping in view of these objectives, teaching and research works are carried out in the university.

AIM:

This paper aims to propose a gender prediction method that integrates social media users' sentiment. This method is used to predict the gender attributes of users, to realize their search on the construction method of social media user portraits. Previous studies on gender prediction have done little analysis of sentiments. This paper has used the idea of transfer learning to analyze users' sentiments and integrated sentimental features into the existing machine learning, and hence, shows superior performance in terms of accuracy as compared to other methods. This paper mainly uses machine learning methods to realize the construction of social media user profiling. Gender attributes are studied. Firstly, feature extraction is carried out for text data of media users. Then the idea of transfer learning is used to analyze the user's sentiment and integrate the sentiment characteristics into the existing machine learning. Finally, five prediction methods, i.e., Logistic Regression (LR), Naïve Bayes (NB), k-Nearest Neighbor (KNN), Random Forest (RF) and Support Vector Machines (SVM) are used to predict the gender of the fused sentiments. The results show that the gender prediction effect after sentiment fusion is better than that before. and the accuracy is increased by about 2.1% on average.

EXISTING SYSTEM:

The construction of user profiling method is the process of classifying users according to established categories through a series of data mining methods. It can maximize the value and improve the team decision-making efficiency, so that users can improve the efficiency of information acquisition and accurately meet their own needs for product applications. The purpose of user profiling is to make a profiling of different dimensions of users and extract and label information such as users' demographic characteristics, social relations, behavioral patterns, habit preferences andideo logical views. In recent years, with the rapid, large-scale and full-coverage development of the Internet, sina Weibo (social media like Twitter, Facebook, Sina Weibo) has become increasingly popular as a social network platform. It has the characteristics of a large user group, fast news transmission speed, wide influence and group effect. Advertising media and social public opinion supervision departments urgently need to dig out accurate and usable information through the attribute analysis of weibo users.

DISADVANTAGES OF EXISTING SYSTEM:

- prediction result is not accurate.
- The above research did not consider the sentimental differences of users of different genders, and users' sentiments can have an impact on user gender prediction.

Algorithm: Logistic Regression

PROPOSED SYSTEM:

This paper adopts the use of K-nearest neighbor for gender prediction based on tolerance rough set method, but this method is susceptible to tolerance threshold. ZHANG Pu,et al [12] used two methods, manual feature construction classifier and convolutional neural network model, were used to predict the gender of users. The results of the two classifiers were fused with the XGBoost model to get the final prediction results. CAOYang, for the issue that asking let extdata cannot accurately predict the gender of inactive users on Weibo, the base classifier was constructed by jointly learning the original microblog data, microblog username data and microblog source data, and then the base classifier was fused by Bayes to finally infer the gender of users. the method of standardization based on maximum and minimum values. Weibo data used does not contain sentimental labels, so the LSTM (Long short-term Memory) neural network is trained by using the data with sentimental labels in the product evaluation of online shopping platform, and then transferred to learn the prediction of sentimental characteristics of Weibo users of different genders. Figure1 is the transfer learning method used in the sentimental learning process.

ADVANTAGES OF PROPOSED SYSTEM:

- The confusion matrix and accuracy of preliminary prediction before and after data adjustment are shown data is a comparison experiment based on the prediction method of Logistic Regression.
- The efficiency and suitability of the proposed approach are compared with other suggested methods.

Algorithm: Machine learning, Classification Technique, supervised machine learning.

SYSTEM REQUIREMENTS:

HARDWARE REQUIREMENTS:

System	:	Intel i5 6 core.
Hard Disk	:	500 GB SSD.
Monitor	:	15'' LED
Input Devices	:	Keyboard, Mouse
Ram	:	32 GB.

SOFTWARE REQUIREMENTS:

Operating system	:	Windows 10.
Coding Language	:	Python
Tool	:	PyCharm, Visual Studio Code
Database	:	SQLite

SYSTEM ANALYSIS

EXISTING SYSTEM:

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REQUIREMENT ANALYSIS

The project involved analyzing the design of a few applications to make the application more user friendly. To do so, it was important to keep the navigations from one screen to the other well-ordered and at the same time reduce the amount of typing the user needs to do. To make the application more accessible, the browser version had to be chosen so that it is compatible with most of the Browsers.

REQUIREMENT SPECIFICATION

Functional Requirements

- Graphical User interface with the User.

Software Requirements

For developing the application, the following are the Software Requirements:

1. Python

2. Django

Operating Systems supported

1. Windows 10 64-bit OS

Technologies and Languages used to Develop

1. Python

Debugger and Emulator

- Any Browser (Particularly Chrome)

Hardware Requirements

For developing the application, the following are the Hardware Requirements:

- Processor: Intel i9
- RAM: 32 GB
- Space on Hard Disk: minimum 1 TB

SYSTEM STUDY

FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and the business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are, [OBJ]

- ◆ **ECONOMICAL FEASIBILITY**
- ◆ **TECHNICAL FEASIBILITY**
- ◆ **SOCIAL FEASIBILITY**

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of funds that the company can pour into the research and development of the system is limited. The expenditure must be justified. Thus, the developed system is well within the budget, and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

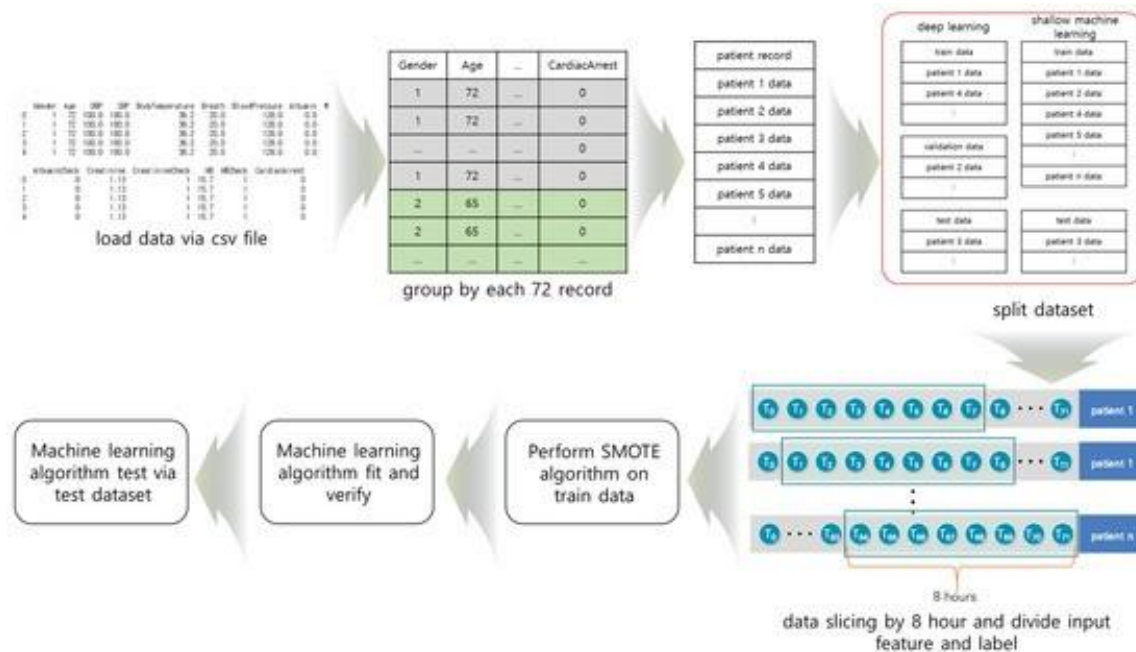
This study is carried out to check the technical feasibility, that is, the technical requirements of the system. No system developed must not have a high demand for the available technical resources. This will lead to high demand for the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

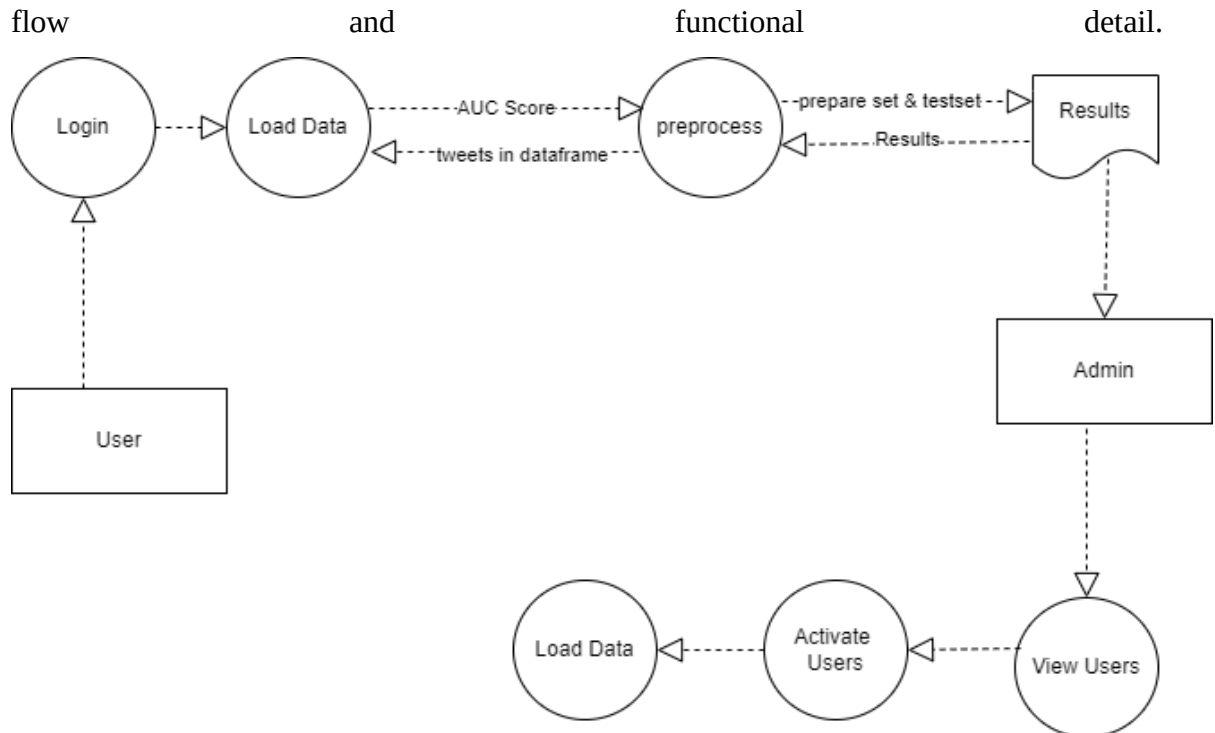
SYSTEM DESIGN

SYSTEM ARCHITECTURE:



DATA FLOW DIAGRAM:

1. The DFD is also called bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.
2. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.
3. DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.
4. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information



UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

UML is a very important part of developing objects-oriented software and the software development process. UML uses mostly graphical notations to express the design of software projects.

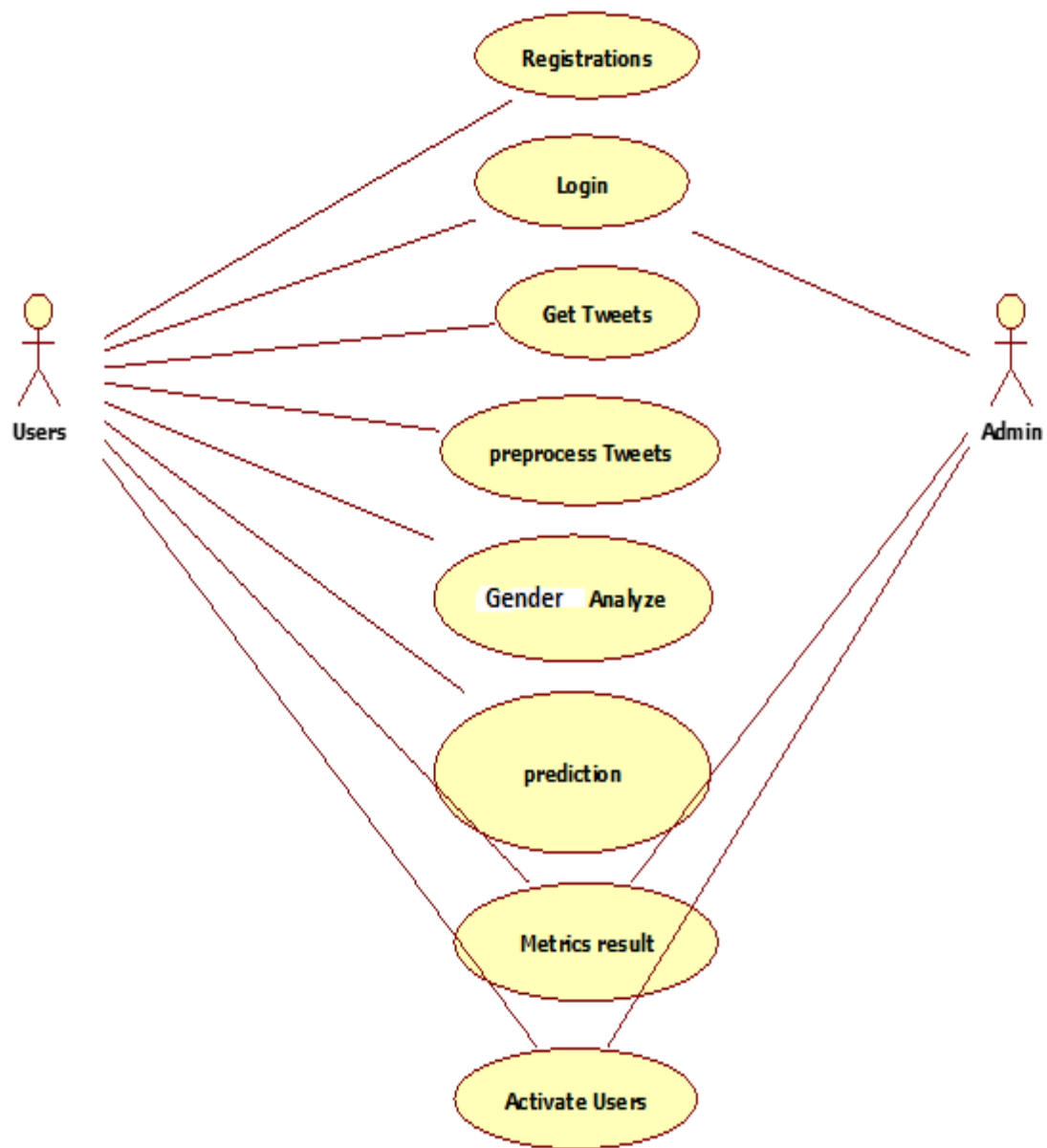
GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns, and components.
7. Integrate best practices.

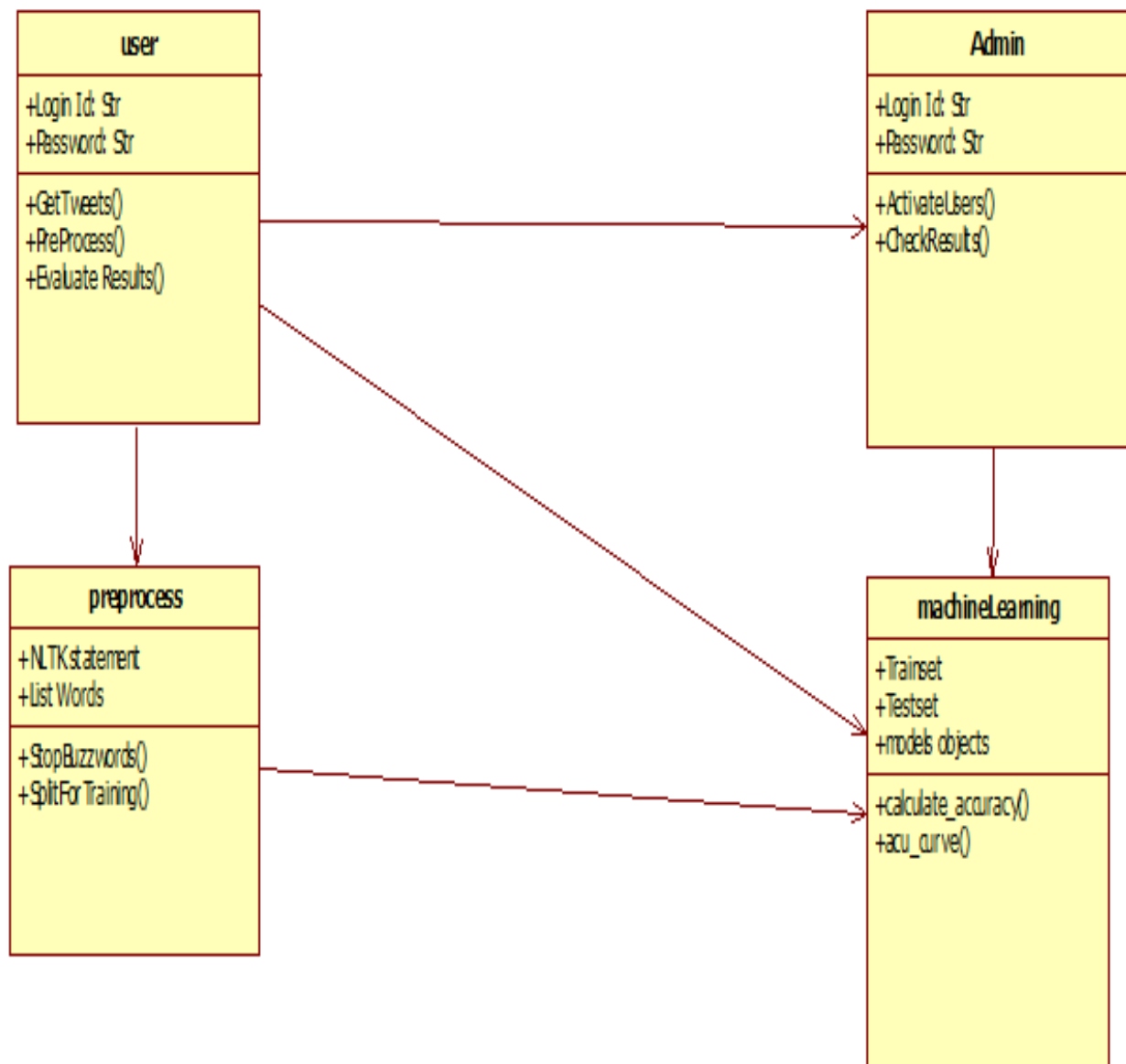
USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.



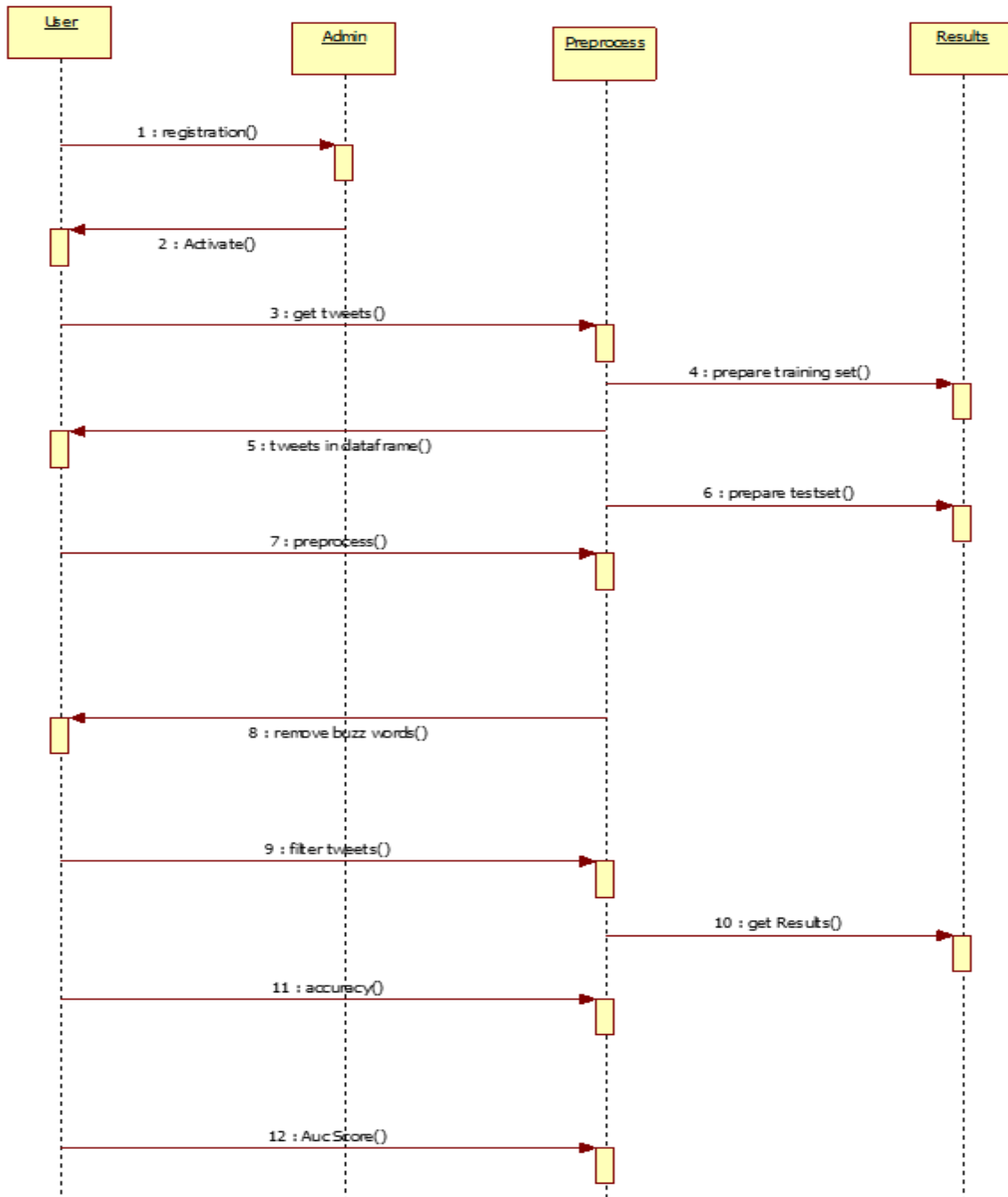
CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.



SEQUENCE DIAGRAM:

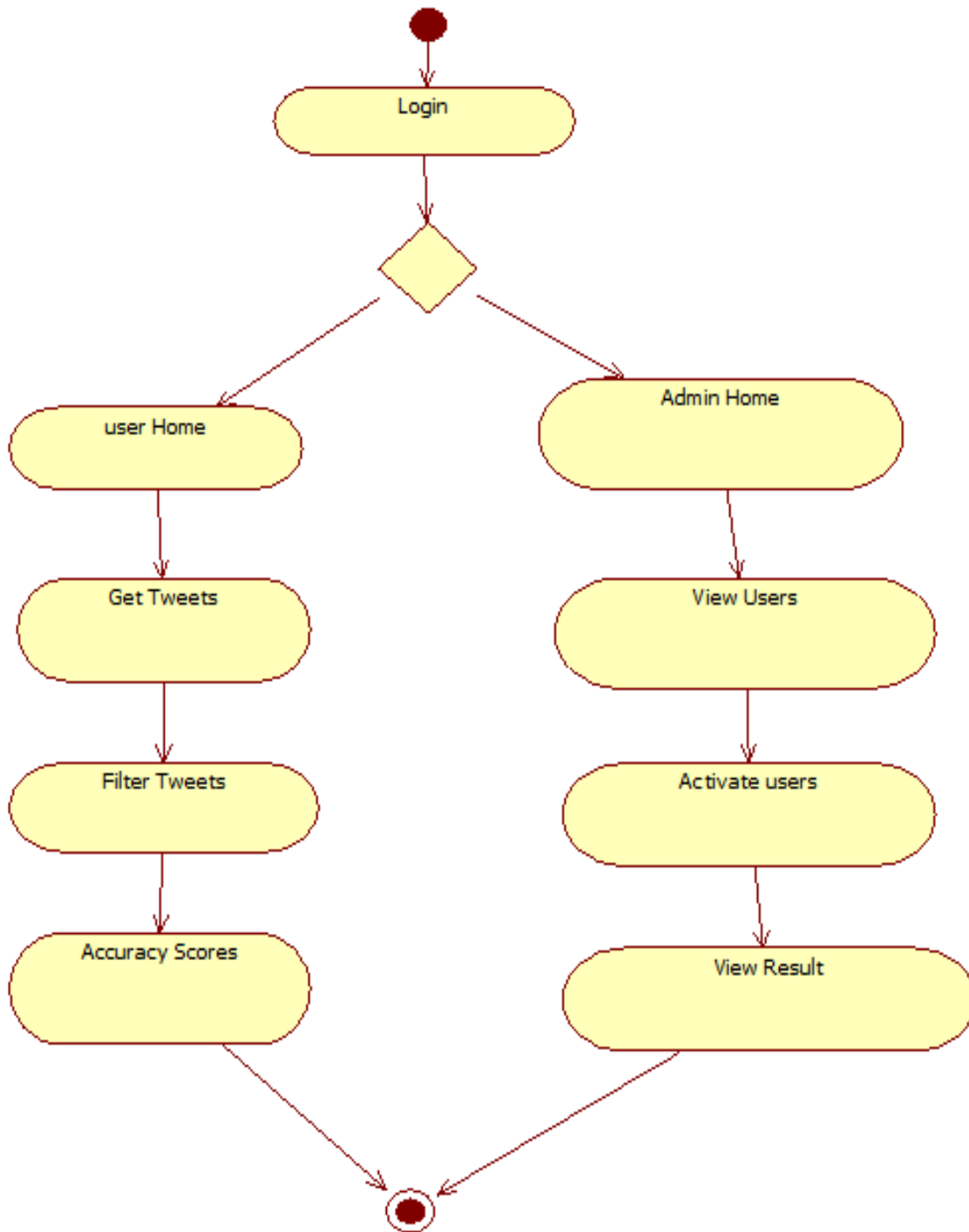
A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.



ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration, and concurrency. In the Unified Modeling Language, activity

diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.



INPUT AND OUTPUT DESIGN

INPUT DESIGN

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data into a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way that it provides security and ease of use while retaining privacy. Input Design considered the following things:

- What data should be given as input?
- How should the data be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when errors occur.

OBJECTIVES

1. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.

2. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulations can be performed. It also provides record viewing facilities.

3. When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided when needed so that the user will not be in a maze of instant. Thus, the objective of input design is to create an input layout that is easy to follow

OUTPUT DESIGN

A quality output is one which meets the requirements of the end user and presents the information clearly. In any system the results of processing are communicated to the users and to other systems through outputs. In output design it is determined how the information is to be displayed for immediate need and the hard copy output. It is the most important and direct source of information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can be used easily and effectively. When analysing design computer output, they should identify the specific output that is needed to meet the requirements.

2. Select methods for presenting information.

3. Create documents, reports, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- Convey information about past activities, current status or projections of the
- Future.
- Signal important events, opportunities, problems, or warnings.
- Trigger an action.
- Confirm an action.

SYSTEM TEST

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedure: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a test in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is the purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, like most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a test in which the software under test is treated as a black box. you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually, and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g., components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Sample Test cases:

S.no	Test cases	Expected Results	Result	Remarks (IF FAILS)
1.	User Register	If user Registration successfully	pass	If already user email exists, then it fails.
2.	User Login	If Username and password is correct, then it will getting valid page.	Pass	Un Register Users will not log in.
3.	Admin login	Admin can login with his login credential. If success he gets his home page	Pass	Invalid login details will not allow here
4.	Admin can activate the register users	Admin can activate the register user id	Pass	If user id not found, then it won't login.
5.	Admin can view machine learning results	Admin can view Algorithm, Accuracy and Area Under Curve Values	Pass	If dataset not found, then it failed
6.	User Can view DataSet	So, information about gender and text is viewed	Pass	If User is not logged in, you cannot view Dataset
7	Calculate the accuracy	Accuracy must be 0 to100 % based on wrong and correct	pass	The accuracy calculated

		prediction it will calculate		
8	User can view machine learning results	User can view Algorithm, Accuracy and Area Under Curve Values	Pass	If dataset not found, then it failed
9	User can view Gender Predict Form	User can write some tweet and predict the gender based on dataset.	Pass	If dataset not found, then it failed
10	Finding Gender	We will load the model and user written test given as test input, based on training model it will predict	pass	Gender predicated

IMPLEMENTATION:

MODULES:

- UserAdmin
- Data Preprocessing
- Machine Learning

MODULES DESCRIPTION:

User: The User can register first. While registering he required a valid user email and mobile for further communications. Once the user registers then admin can activate the user. Once admin activated the user then user can login into our system. Users can view his homepage. The User can view Dataset which collected Kaggle twitter dataset and applying the machine learning technique to the dataset and finding accuracy and auc. The results displayed in the browser

Admin: Admin can login with his login details. Admin can activate the registered users. He can view Users Details like Name, Login Id, Mobile, Email, Locality, status Once he activates then only the user can login into our system. Admin can view machine learning results and can check Algorithm, Accuracy and Area Under Curve. Admin can view the overall user data in the browser. Once his work is done, he can log out.

Data Preprocessing: A dataset can be viewed as a collection of data objects, which are often also called records, points, vectors, patterns, events, cases, samples, observations, or entities. Data objects are described by a few features that capture the basic characteristics of an object, such as the mass of a physical object or the time at which an event occurred, etc. Features are often called as variables, characteristics, fields, attributes, or dimensions. The expulsion of missing information, modifying default values if relevant and grouping of attributes for prediction at various levels.

Machine learning: The dataset is subjected to five machine learning classifiers such as Naïve Bayes (NB), K-Nearest-Neighbor (KNN), Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM). The classifier which bags up the highest accuracy could be determined as the best classifier. For all algorithms, accuracy and area under curve is calculated and displayed in results.

CONCLUSION

This paper mainly studies the prediction of social media user profiling. Firstly, starting from the background knowledge, related work and target characteristics of social media user profiling, expounds the purpose and significance of studying social media user profiling. Then the idea of transfer learning is used to analyze the user's sentiment and integrate the sentiment characteristics into the existing machine learning. Finally, five prediction methods, i.e., LR, NB, KNN, RF and SVM, were used to predict the gender of fused sentiments.

Future Enhancement

The prediction of social media user profiling in this paper is based on the gender attributes of social media users. Although certain effects have been achieved, many other attributes of social media users will only serve as a starting point. Although the research ideas and experimental methods in this paper have certain refer ability, there are still many inconsiderate places in the whole experiment process, and many specific related factors need to be further sorted out and analyzed.

SCREEN SHOTS

Home Page:

User Register Form

Research on Gender Prediction for Social Media User Profiling by machine learning method

Home User Admin Registration

17:47 05-01-2022

User Register Form

User Name

Login ID

Password

Mobile

email

Locality

Address

City

State

18:02 05-01-2022

Admin Home Page

Admin Login Form

• •

Admin Activation page



User Home page

User Login Form

•

•

View Tweets

•

Gender Predict Form

View Results

The screenshot displays a web application interface. At the top, there are two tabs: "Gender Predict Form" and "View Results". The "View Results" tab is active, showing a table of machine learning results. Below the table, there is a section titled "Gender Predict Form....." which includes a text input field and a "Predict" button. The browser's address bar shows the URL "192.168.1.67:8000/user_machine_learning/". The Windows taskbar at the bottom shows the time as 18:25 on 05-01-2022.

Machine Learning Results.....

S.NO	Algorithm	Accuracy	Area Under Curve
1	Navie Bayes	0.5829145728643216	0.5117605131860299
2	KNN	0.5678391959798995	0.491743882157282
3	Logistic Regression	0.6331658291457286	0.5159776669042528
4	Random Forest	0.6482412060301508	0.5634354953670706
5	Support Vector Machine	0.592964824120603	0.5007127583749109

Gender Predict Form.....

Tweet: I am Coding Written by **Male**
[Write Some Tweet Here](#)

Predict

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SOURCE CODE

user side View.py

```
from django.shortcuts import render,HttpResponse
from django.contrib import messages
from .forms import UserRegistrationForm
from .models import UserRegistrationModel
import pandas as pd

# Create your views here.

def UserRegisterActions(request):
    if request.method == 'POST':
        form = UserRegistrationForm(request.POST)
        if form.is_valid():
            print('Data is Valid')
            form.save()
            messages.success(request, 'You have been successfully registered')
            form = UserRegistrationForm()
            return render(request, 'UserRegistrations.html', {'form': form})
        else:
            messages.success(request, 'Email or Mobile Already Existed')
            print("Invalid form")
    else:
        form = UserRegistrationForm()
    return render(request, 'UserRegistrations.html', {'form': form})

def UserLoginCheck(request):
    if request.method == "POST":
        loginid = request.POST.get('loginid')
        pswd = request.POST.get('pswd')
        print("Login ID = ", loginid, ' Password = ', pswd)
        try:
            check = UserRegistrationModel.objects.get(loginid=loginid, password=pswd)
            status = check.status
            print('Status is = ', status)
            if status == "activated":
                request.session['id'] = check.id
                request.session['loggeduser'] = check.name
                request.session['loginid'] = loginid
                request.session['email'] = check.email
                print("User id At", check.id, status)
                return render(request, 'users/UserHomePage.html', {})
            else:
```

```

        messages.success(request, 'Your Account Not at activated')
        return render(request, 'UserLogin.html')
    except Exception as e:
        print('Exception is ', str(e))
        pass
    messages.success(request, 'Invalid Login id and password')
    return render(request, 'UserLogin.html', {})

```

```

def UserHome(request):
    return render(request, 'users/UserHomePage.html', {})

```

```

def gender_dataset(request):
    from django.conf import settings
    import os
    path = os.path.join(settings.MEDIA_ROOT, "gender_dataset.csv")
    pd.options.display.max_colwidth=300
    df = pd.read_csv(path,nrows=1000)
    df = df[['gender','text']]
    print(df.head())
    df = df.to_html()
    #print(df)
    return render(request, 'users/tweetsdata.html',{'data': df})

```

```

def user_machine_learning(request):
    from .utility import process_classification
    nb_accuracy,nb_cm,nb_auc = process_classification.calculate_naive_bayes()
    knn_accuracy, knn_cm, knn_auc = process_classification.calculate_knn()
    lg_accuracy, lg_cm, lg_auc = process_classification.calculate_logistic_regression()
    rf_accuracy, rf_cm, rf_auc = process_classification.calculate_random_forest()
    svm_accuracy, svm_cm, svm_auc = process_classification.calculate_svm()
    nb = {'nb_accuracy': nb_accuracy, 'nb_auc': nb_auc}
    knn = {'knn_accuracy': knn_accuracy, 'knn_auc': knn_auc}
    lg = {'lg_accuracy': lg_accuracy, 'lg_auc': lg_auc}
    rf = {'rf_accuracy': rf_accuracy, 'rf_auc': rf_auc}
    svm = {'svm_accuracy': svm_accuracy, 'svm_auc': svm_auc}
    return render(request, 'users/user_classification.html', {"nb":
nb,"knn":knn,"lg":lg,"rf":rf,"svm": svm})

```

```

def user_gender_predict(request):
    if request.method=="POST":
        tweet = request.POST.get('text')
        from .utility import process_classification
        rslt = process_classification.process_user_tweet(tweet)

```

```

print(f"{rslt} and type {type(rslt)}")
if rslt== 1.0:
    gender = "Male"
else:
    gender = "Female"
return render(request, "users/user_gender_predict_form.html", {"gender": gender,
"tweet": tweet})
else:
    return render(request, "users/user_gender_predict_form.html",{})

```

Users models.py

```

from django.db import models

```

Create your models here.

```

class UserRegistrationModel(models.Model):
    name = models.CharField(max_length=100)
    loginid = models.CharField(unique=True, max_length=100)
    password = models.CharField(max_length=100)
    mobile = models.CharField(unique=True, max_length=100)
    email = models.CharField(unique=True, max_length=100)
    locality = models.CharField(max_length=100)
    address = models.CharField(max_length=1000)
    city = models.CharField(max_length=100)
    state = models.CharField(max_length=100)
    status = models.CharField(max_length=100)

```

```

def __str__(self):
    return self.loginid

```

```

class Meta:
    db_table = 'UserRegistrations'

```

user side forms.py

```

from django import forms
from .models import UserRegistrationModel

```

```

class UserRegistrationForm(forms.ModelForm):
    name = forms.CharField(widget=forms.TextInput(attrs={'pattern': '[a-zA-Z]+'}),
required=True, max_length=100)
    loginid = forms.CharField(widget=forms.TextInput(attrs={'pattern': '[a-zA-Z]+'}),
required=True, max_length=100)
    password = forms.CharField(widget=forms.PasswordInput(attrs={'pattern': '(?=.*\d)(?=.*[a-
z])(?=.*[A-Z]).{8,}'},

```

```

        'title': 'Must contain at least one number and one
uppercase and lowercase letter, and at least 8 or more characters'})),
        required=True, max_length=100)
    mobile = forms.CharField(widget=forms.TextInput(attrs={'pattern': '[56789][0-9]{9}'}),
required=True,
        max_length=100)
    email = forms.CharField(widget=forms.TextInput(attrs={'pattern': '[a-z0-9._%+-]+@[a-z0-
9.-]+\.[a-z]{2,}$'})),
        required=True, max_length=100)
    locality = forms.CharField(widget=forms.TextInput(), required=True, max_length=100)
    address = forms.CharField(widget=forms.Textarea(attrs={'rows': 4, 'cols': 22}),
required=True, max_length=250)
    city = forms.CharField(widget=forms.TextInput(
        attrs={'autocomplete': 'off', 'pattern': '[A-Za-z ]+', 'title': 'Enter Characters Only '}),
required=True,
        max_length=100)
    state = forms.CharField(widget=forms.TextInput(
        attrs={'autocomplete': 'off', 'pattern': '[A-Za-z ]+', 'title': 'Enter Characters Only '}),
required=True,
        max_length=100)
    status = forms.CharField(widget=forms.HiddenInput(), initial='waiting', max_length=100)

class Meta():
    model = UserRegistrationModel
    fields = '__all__'

```

Process Classification.py

```

from django.conf import settings
import os
import pandas as pd
from sklearn import metrics
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.svm import LinearSVC
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier

def clean_dataset(df):
    assert isinstance(df, pd.DataFrame), df.dropna(inplace=True)

```

```

path = os.path.join(settings.MEDIA_ROOT, "gender_dataset.csv")
pd.options.display.max_colwidth=300
# df = pd.read_csv(path,nrows=1000)
data = pd.read_csv(path,nrows=1000)
data = data[['gender', 'text']]

print(data['gender'].unique())
data['gender'].replace('female',0, inplace=True)
data['gender'].replace('male', 1,inplace=True)
data['gender'].replace('brand',1, inplace=True)
data['gender'].replace('unknown',0, inplace=True)
data.dropna(inplace=True)
#print(data.head())
X = data.text
y = data.gender
print("=====>",y)
#Using CountVectorizer to convert text into tokens/features
vect = CountVectorizer(stop_words='english', ngram_range = (1,1), max_df = .80, min_df = 4)
X_train, X_test, y_train, y_test = train_test_split(X,y,random_state=1, test_size= 0.2)
#Using training data to transform text into counts of features for each message
vect.fit(X_train)
X_train_dtm = vect.transform(X_train)
X_test_dtm = vect.transform(X_test)

def calculate_naive_bayes():
    # Accuracy using Naive Bayes Model
    NB = MultinomialNB()
    NB.fit(X_train_dtm, y_train)
    y_pred = NB.predict(X_test_dtm)
    print('\nNaive Bayes')
    nb_accuracy = metrics.accuracy_score(y_test, y_pred)
    nb_cm = metrics.confusion_matrix(y_test, y_pred)
    nb_auc = metrics.roc_auc_score(y_test, y_pred)
    return nb_accuracy,nb_cm,nb_auc

def calculate_knn():
    KNN = KNeighborsClassifier()
    KNN.fit(X_train_dtm, y_train)
    y_pred = KNN.predict(X_test_dtm)
    print('\n K Nearest Neighbour')
    knn_accuracy = metrics.accuracy_score(y_test, y_pred)
    knn_cm = metrics.confusion_matrix(y_test, y_pred)
    knn_auc = metrics.roc_auc_score(y_test, y_pred)
    return knn_accuracy, knn_cm, knn_auc

```

```
def calculate_logistic_regression():
    lg = LogisticRegression()
    lg.fit(X_train_dtm, y_train)
    y_pred = lg.predict(X_test_dtm)
    print('\n Logistic Regression')
    lg_accuracy = metrics.accuracy_score(y_test, y_pred)
    lg_cm = metrics.confusion_matrix(y_test, y_pred)
    lg_auc = metrics.roc_auc_score(y_test, y_pred)
    return lg_accuracy, lg_cm, lg_auc
```

```
def calculate_random_forest():
    rf = RandomForestClassifier()
    rf.fit(X_train_dtm, y_train)
    y_pred = rf.predict(X_test_dtm)
    print('\n Random Forest')
    rf_accuracy = metrics.accuracy_score(y_test, y_pred)
    rf_cm = metrics.confusion_matrix(y_test, y_pred)
    rf_auc = metrics.roc_auc_score(y_test, y_pred)
    return rf_accuracy, rf_cm, rf_auc
```

```
def calculate_svm():
    SVM = LinearSVC()
    SVM.fit(X_train_dtm, y_train)
    y_pred = SVM.predict(X_test_dtm)
    print('\nSupport Vector Machine')
    svm_accuracy = metrics.accuracy_score(y_test, y_pred)
    svm_cm = metrics.confusion_matrix(y_test, y_pred)
    svm_auc = metrics.roc_auc_score(y_test, y_pred)
    return svm_accuracy, svm_cm, svm_auc
```

```
def process_user_tweet(tweet):
    # Custom Test: Test a review on the best performing model (Logistic Regression)
    trainingVector = CountVectorizer(stop_words='english', ngram_range=(1, 1), max_df=.80,
min_df=5)
    trainingVector.fit(X)
    X_dtm = trainingVector.transform(X)
    LR_complete = LogisticRegression()
    LR_complete.fit(X_dtm, y)
    # Input Review
    test = []
    test.append(tweet)
```

```

test_dtm = trainingVector.transform(test)
predLabel = LR_complete.predict(test_dtm)
tags = [0., 1.]
# Display Output
return tags[int(predLabel[0])]

```

Admin based Views.py

```

from django.http.response import HttpResponseRedirect
from django.shortcuts import render
from django.contrib import messages
from users.models import UserRegistrationModel
import pandas as pd

# Create your views here.
def AdminLoginCheck(request):
    if request.method == 'POST':
        usrid = request.POST.get('loginid')
        pswd = request.POST.get('pswd')
        print("User ID is = ", usrid)
        if usrid == 'admin' and pswd == 'admin':
            return render(request, 'admins/AdminHome.html')

    else:
        messages.success(request, 'Please Check Your Login Details')
    return render(request, 'AdminLogin.html', {})

def AdminHome(request):
    return render(request, 'admins/AdminHome.html')

def RegisterUsersView(request):
    data = UserRegistrationModel.objects.all()
    return render(request, 'admins/viewregisterusers.html', {'data':data})

def ActivaUsers(request):
    if request.method == 'GET':
        id = request.GET.get('uid')
        status = 'activated'
        print("PID = ", id, status)
        UserRegistrationModel.objects.filter(id=id).update(status=status)
        data = UserRegistrationModel.objects.all()
        return render(request, 'admins/viewregisterusers.html', {'data':data})

```

```

def admin_view_results(request):
    from users.utility import process_classification
    nb_accuracy, nb_cm, nb_auc = process_classification.calculate_naive_bayes()
    knn_accuracy, knn_cm, knn_auc = process_classification.calculate_knn()
    lg_accuracy, lg_cm, lg_auc = process_classification.calculate_logistic_regression()
    rf_accuracy, rf_cm, rf_auc = process_classification.calculate_random_forest()
    svm_accuracy, svm_cm, svm_auc = process_classification.calculate_svm()
    nb = {'nb_accuracy': nb_accuracy, 'nb_auc': nb_auc}
    knn = {'knn_accuracy': knn_accuracy, 'knn_auc': knn_auc}
    lg = {'lg_accuracy': lg_accuracy, 'lg_auc': lg_auc}
    rf = {'rf_accuracy': rf_accuracy, 'rf_auc': rf_auc}
    svm = {'svm_accuracy': svm_accuracy, 'svm_auc': svm_auc}
    return render(request, 'admins/admin_results.html', {"nb": nb, "knn": knn, "lg": lg, "rf":
rf, "svm": svm})

```

Userbase.py

```

{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="utf-8">
    <meta content="width=device-width, initial-scale=1.0" name="viewport">

    <title>Project</title>
    <meta content="" name="description">
    <meta content="" name="keywords">

    <!-- Favicons -->
    <link href="{% static 'img/favicon.png' %}" rel="icon">
    <link href="{% static 'img/apple-touch-icon.png' %}" rel="apple-touch-icon">

    <!-- Google Fonts -->
    <link
href="https://fonts.googleapis.com/css?family=Open+Sans:300,300i,400,400i,600,600i,700,7
00i|Raleway:300,300i,400,400i,500,500i,600,600i,700,700i|Poppins:300,300i,400,400i,500,500
i,600,600i,700,700i" rel="stylesheet">

    <!-- Vendor CSS Files -->
    <link href="{% static 'vendor/aos/aos.css' %}" rel="stylesheet">
    <link href="{% static 'vendor/bootstrap/css/bootstrap.min.css' %}" rel="stylesheet">
    <link href="{% static 'vendor/bootstrap-icons/bootstrap-icons.css' %}" rel="stylesheet">

```

```

<link href="{% static 'vendor/boxicons/css/boxicons.min.css' %}" rel="stylesheet">
<link href="{% static 'vendor/glightbox/css/glightbox.min.css' %}" rel="stylesheet">
<link href="{% static 'vendor/swiper/swiper-bundle.min.css' %}" rel="stylesheet">

<!-- Template Main CSS File -->
<link href="{% static 'css/style.css' %}" rel="stylesheet">

<!--
=====
* Template Name: Bocor - v4.5.0
* Template URL: https://bootstrapmade.com/bocor-bootstrap-template-nice-animation/
* Author: BootstrapMade.com
* License: https://bootstrapmade.com/license/
=====
-->
</head>

<body>

<!-- ===== Header ===== -->
<header id="header">
<div class="container d-flex align-items-center justify-content-between">

<div class="logo">
<h1><a href="index.html"><span>Machine Learning</span></a></h1>
<!-- Uncomment below if you prefer to use an image logo -->
<!-- <a href="index.html"></a>-->
</div>

<nav id="navbar" class="navbar">
<ul>
<li><a class="nav-link scrollto" href="{% url 'UserHome' %}">Home</a></li>
<li><a class="nav-link scrollto" href="{% url 'gender_dataset' %}">View
Tweets</a></li>
<li><a class="nav-link scrollto" href="{% url 'user_machine_learning' %}">Machine Learning</a></li>
<li><a class="nav-link scrollto" href="{% url 'user_gender_predict' %}">Prediction</a></li>
<li><a class="nav-link scrollto" href="{% url 'index' %}">Logout</a></li>

<!-- <li><a class="nav-link scrollto" href="#team">Team</a></li>-->
<!-- <li class="dropdown"><a href="#"><span>Drop Down</span> <i class="bi bi-chevron-down"></i></a>-->
<!-- <ul>-->
<!-- <li><a href="#">Drop Down 1</a></li>-->

```

```

<!--      <li class="dropdown"><a href="#"><span>Deep Drop Down</span> <i class="bi
bi-chevron-right"></i></a>-->
<!--      <ul>-->
<!--      <li><a href="#">Deep Drop Down 1</a></li>-->
<!--      <li><a href="#">Deep Drop Down 2</a></li>-->
<!--      <li><a href="#">Deep Drop Down 3</a></li>-->
<!--      <li><a href="#">Deep Drop Down 4</a></li>-->
<!--      <li><a href="#">Deep Drop Down 5</a></li>-->
<!--      </ul>-->
<!--      </li>-->
<!--      <li><a href="#">Drop Down 2</a></li>-->
<!--      <li><a href="#">Drop Down 3</a></li>-->
<!--      <li><a href="#">Drop Down 4</a></li>-->
<!--      </ul>-->
<!--      </li>-->
<!--      <li><a class="nav-link scrollto" href="#contact">Contact</a></li>-->
<!--      <li><a class="getstarted scrollto" href="#about">Get Started</a></li>-->
<!--      </ul>-->
      <i class="bi bi-list mobile-nav-toggle"></i>
    </nav><!-- .navbar -->

  </div>
</header><!-- End Header -->

<main id="main">
  <!-- ===== Breadcrumbs Section ===== -->
  <section class="breadcrumbs">
    <div class="container">

      <div class="d-flex justify-content-between align-items-center">
        <h2 style="font-weight:bold;">Research on Gender Prediction for Social Media User
        Profiling by machine learning method</h2>
      <!--      <ol>-->
      <!--      <li><a href="index.html">Home</a></li>-->
      <!--      <li>Inner Page</li>-->
      <!--      </ol>-->
    </div>

  </div>
</section><!-- End Breadcrumbs Section -->

  <section class="inner-page">
    <div class="container">

      {%block contents%}

```

```

{%endblock%}

</div>
</section>

</main><!-- End #main -->

<!-- ===== Footer ===== -->
<footer id="footer">

  <div class="footer-top">

    <div class="container">

      </div>
</div>

<div class="container footer-bottom clearfix">
  <div class="copyright">
    &copy; Copyright <strong><span>2021 Alex Corporation</span></strong>. All Rights
Reserved
  </div>
  <div class="credits">
    <!-- All the links in the footer should remain intact. -->
    <!-- You can delete the links only if you purchased the pro version. -->
    <!-- Licensing information: https://bootstrapmade.com/license/ -->
    <!-- Purchase the pro version with working PHP/AJAX contact form:
https://bootstrapmade.com/bocor-bootstrap-template-nice-animation/ -->
    <!-- Designed by <a href="https://bootstrapmade.com/">BootstrapMade</a> -->
  </div>
</div>
</footer><!-- End Footer -->

<a href="#" class="back-to-top d-flex align-items-center justify-content-center"><i
class="bi bi-arrow-up-short"></i></a>

<!-- Vendor JS Files -->
<script src="{% static 'vendor/aos/aos.js' %}"></script>
<script src="{% static 'vendor/bootstrap/js/bootstrap.bundle.min.js' %}"></script>
<script src="{% static 'vendor/glightbox/js/glightbox.min.js' %}"></script>
<script src="{% static 'vendor/isotope-layout/isotope.pkgd.min.js' %}"></script>
<script src="{% static 'vendor/php-email-form/validate.js' %}"></script>
<script src="{% static 'vendor/swiper/swiper-bundle.min.js' %}"></script>

```

```

<!-- Template Main JS File -->
<script src="{% static 'js/main.js' %}"></script>

</body>
</html>
index.html

{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="utf-8">
  <meta content="width=device-width, initial-scale=1.0" name="viewport">

  <title>Project</title>
  <meta content="" name="description">
  <meta content="" name="keywords">

  <!-- Favicons -->
  <link href="{% static 'img/favicon.png' %}" rel="icon">
  <link href="{% static 'img/apple-touch-icon.png' %}" rel="apple-touch-icon">

  <!-- Google Fonts -->
  <link
href="https://fonts.googleapis.com/css?family=Open+Sans:300,300i,400,400i,600,600i,700,700i|Raleway:300,300i,400,400i,500,500i,600,600i,700,700i|Poppins:300,300i,400,400i,500,500i,600,600i,700,700i" rel="stylesheet">

  <!-- Vendor CSS Files -->
  <link href="{% static 'vendor/aos/aos.css' %}" rel="stylesheet">
  <link href="{% static 'vendor/bootstrap/css/bootstrap.min.css' %}" rel="stylesheet">
  <link href="{% static 'vendor/bootstrap-icons/bootstrap-icons.css' %}" rel="stylesheet">
  <link href="{% static 'vendor/boxicons/css/boxicons.min.css' %}" rel="stylesheet">
  <link href="{% static 'vendor/glightbox/css/glightbox.min.css' %}" rel="stylesheet">
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</head>

<body>

<!-- ===== Header ===== -->
<header id="header">
  <div class="container d-flex align-items-center justify-content-between">

    <div class="logo">
      <h1><a href="index.html"><span>Machine Learning</span></a></h1>
      <!-- Uncomment below if you prefer to use an image logo -->
      <!-- <a href="index.html"></a>-->
    </div>

    <nav id="navbar" class="navbar">
      <ul>
        <li><a class="nav-link scrollto active" href="{% url 'index' %}">Home</a></li>
        <li><a class="nav-link scrollto" href="{% url 'UserLogin' %}">User</a></li>
        <li><a class="nav-link scrollto" href="{% url 'AdminLogin' %}">Admin</a></li>
        <li><a class="nav-link scrollto" href="{% url 'UserRegister' %}">Registration</a></li>

      </ul>
      <i class="bi bi-list mobile-nav-toggle"></i>
    </nav><!-- .navbar -->

  </div>
</header><!-- End Header -->

<!-- ===== Hero Section ===== -->
<section id="hero">

  <div class="container">
    <div class="row d-flex align-items-center">
      <div class="col-lg-6 py-5 py-lg-0 order-2 order-lg-1" data-aos="fade-right">
        <h1>Research on Gender Prediction for Social Media User Profiling by machine learning method</h1>
        <!-- <h2>We are team of talented designers making websites with Bootstrap</h2>-->
        <!-- <a href="#about" class="btn-get-started scrollto">Get Started</a>-->
      </div>
      <div class="col-lg-6 order-1 order-lg-2 hero-img" data-aos="fade-left">
        
      </div>
    </div>
  </div>

```

</div>

</div>

</section><!-- End Hero -->

<main id="main">

<!-- <!– ===== Services Section ===== –>-->

<section id="services" class="services section-bg">

<div class="container">

<div class="section-title">

<h2 data-aos="fade-in">Technologies</h

<p data-aos="fade-in" align="justify">

This paper aims to propose a gender prediction method that integrates social media users' sentiment. This method is used to predict the gender attributes of users, so as to realize the research on the construction method of social media user portraits. Previous studies on gender prediction have done little analysis of sentiments. This paper has used the idea of transfer learning to analyse users' sentiments and integrated sentimental features into the existing machine learning, and hence, shows superior performance in terms of accuracy as compared to other methods. This paper mainly uses machine learning method to realize the construction of social media user profiling. The gender attribute is studied. Firstly, feature extraction is carried out for text data of media users. Then the idea of transfer learning is used to Analyse the user's sentiment and integrate the sentiment characteristics into the existing machine learning. Finally, five prediction methods. Logistic Regression (LR), Naïve Bayes (NB), k-Nearest Neighbor (KNN), Random Forest (RF) and Support Vector Machines (SVM) is used to predict the gender of the fused sentiments. The results show that the gender prediction effect after sentiment fusion is better than that before. and the accuracies increased by about 2.1% on average</p>

</div>

<div class="row">

<div class="col-md-6 d-flex align-items-stretch" data-aos="fade-right">

<div class="card">

<div class="card-img">

</div>

<div class="card-body">

<h5 class="card-title">Machine Learning</h5>

<p class="card-text">Machine learning (ML) is a type of artificial intelligence (AI) that allows software applications to become more accurate at predicting outcomes without being explicitly programmed to do so. Machine learning algorithms use historical data as input to predict new output values.</p>

```

<!--      <div class="read-more"><a href="#"><i class="bi bi-arrow-right"></i> Read
More</a></div>-->
    </div>
  </div>
</div>
<div class="col-md-6 d-flex align-items-stretch" data-aos="fade-left">
  <div class="card">
    <div class="card-img">
      
    </div>
    <div class="card-body">
      <h5 class="card-title"><a href="#">Artificial Intelligence</a></h5>
      <p class="card-text">Artificial intelligence is the simulation of human intelligence
processes by machines, especially computer systems. Specific applications of AI include expert
systems, natural language processing, speech recognition and machine vision.</p>
<!--      <div class="read-more"><a href="#"><i class="bi bi-arrow-right"></i> Read
More</a></div>-->
    </div>
  </div>

  </div>
</div>
</div>

</div>
</section><!-- End Services Section -->

</main><!-- End #main -->

<!-- ===== Footer ===== -->
<footer id="footer">

  <div class="footer-top">

    <div class="container text-center">

      <div class="container footer-bottom clearfix" align="center">
        <div class="copyright">
          &copy; Copyright <strong><span>2021 Alex Corporation</span></strong>. All Rights
Reserved
        </div>
        <div class="credits">

```

```

    </div>
</div>
</footer><!-- End Footer -->

<!-- <a href="#" class="back-to-top d-flex align-items-center justify-content-center"><i
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<script src="{% static 'js/main.js' %}"></script>

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```